



Vectors in R Programming

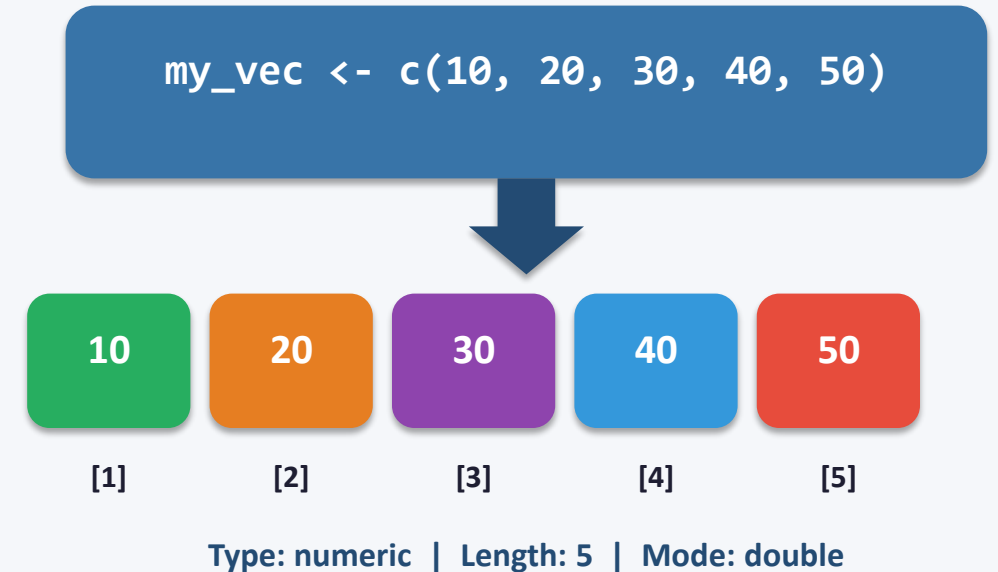
Creation, Types, Operations & Real-World Data Analysis

What is a Vector in R?

- A vector is the **FUNDAMENTAL** data structure in R — everything is a vector!
- A vector is a 1-dimensional array that holds elements of the SAME type
- Created using the `c()` function (combine/concatenate)
- Length: number of elements → `length(vec)`
- Indexing starts at 1 (not 0 like Python/C++)

5 Atomic Vector Types in R:

- numeric : 3.14, 2.5, -1.0 (double-precision floating point)
- integer : 1L, 5L, 100L (whole numbers with 'L' suffix)
- character : "hello", "R" (text/strings)
- logical : TRUE, FALSE (boolean values)
- complex : 3+2i, 1-4i (complex numbers)



```
# Create different vector types
num_vec <- c(1.5, 2.3, 4.7)      # numeric
int_vec  <- c(1L, 2L, 3L)        # integer
char_vec <- c("A", "B", "C")    # character
log_vec  <- c(TRUE, FALSE, TRUE) # logical

# Check type
typeof(num_vec) # "double"
class(int_vec)  # "integer"
```

Step-by-Step: Creating Vectors in R

Step 1: Using `c()` — The Combine Function

The most common way to create vectors.

Combine any values of the same type.

```
ages <- c(25, 30, 35, 40, 45)
names <- c("Alice", "Bob", "Carol")
mixed <- c(1, "two") # coercion to
character
```

Step 2: Using `:` and `seq()` — Sequences

Generate sequences of numbers efficiently.

`:` creates integer sequences; `seq()` is more flexible.

```
v1 <- 1:10 # 1,2,3,...,10
v2 <- seq(0, 100, by=20) # 0,20,40,60,80,100
v3 <- seq(0, 1, length.out=5)
```

Step 3: Using `rep()` — Replicate Values

Repeat values or entire vectors.

Useful for creating labels, factors, patterns.

```
rep(5, 3) # 5, 5, 5
rep(c("A","B"), 3) # A,B,A,B,A,B
rep(1:3, each=2) # 1,1,2,2,3,3
```

Step 4: Vector Operations & Indexing

R performs element-wise operations automatically.

Access elements with `[ ]` brackets.

```
x <- c(10, 20, 30)
x + 5 # 15, 25, 35
x[2] # 20 (2nd element)
x[c(1,3)] # 10, 30
```

Real-World Examples & Applications in R

```
# Monthly sales data (in $1000s)
sales <- c(45, 52, 48, 61, 55, 70)
months <- c("Jan", "Feb", "Mar",
            "Apr", "May", "Jun")

mean(sales)      # 55.17
max(sales)       # 70
sales[sales > 50] # 52, 61, 55, 70
```

```
# Temperature readings (°C)
temps <- c(22.5, 23.1, 21.8, 24.0,
           22.9, 23.5, 21.2)

# Convert to Fahrenheit
f_temps <- temps * 9/5 + 32
# Result: 72.5, 73.6, 71.2, 75.2...

temps[temps > 23] # 23.1, 24.0, 23.5
```

```
# Student exam scores
scores <- c(78, 85, 92, 67, 88, 91, 74)

mean(scores)      # 82.14
sd(scores)        # 9.03
median(scores)    # 85

# Grade assignment
grades <- ifelse(scores >= 90, "A",
                 ifelse(scores >= 80, "B", "C"))
```

Real-World Applications of R Vectors

- Data Science: Feature columns in data frames
- Bioinformatics: Gene expression values, DNA sequences
- Finance: Stock prices, portfolio returns over time
- Healthcare: Patient vital signs, lab results
- Machine Learning: Input features for models
- Survey Analysis: Response ratings, demographic data
- Time Series: Daily measurements, sensor readings

R Vector Functions Cheat Sheet

Function	Description	Example
<code>c()</code>	Combine values into a vector	<code>c(1, 2, 3)</code>
<code>length()</code>	Number of elements	<code>length(vec) → 3</code>
<code>typeof()</code>	Data type of vector	<code>typeof(vec) → "double"</code>
<code>seq()</code>	Generate sequences	<code>seq(1, 10, by=2)</code>
<code>rep()</code>	Replicate values	<code>rep(5, 3) → 5,5,5</code>
<code>sort()</code>	Sort in ascending order	<code>sort(c(3,1,2)) → 1,2,3</code>
<code>rev()</code>	Reverse vector order	<code>rev(c(1,2,3)) → 3,2,1</code>
<code>unique()</code>	Remove duplicates	<code>unique(c(1,1,2)) → 1,2</code>
<code>which()</code>	Find indices matching condition	<code>which(vec > 5)</code>
<code>sum()</code> / <code>mean()</code>	Aggregate functions	<code>sum(vec)</code> , <code>mean(vec)</code>
<code>is.vector()</code>	Check if object is a vector	<code>is.vector(x) → TRUE</code>
<code>as.numeric()</code>	Type coercion	<code>as.numeric(c("1","2"))</code>

💡 **Pro Tip:** Vectors are the building blocks of data frames, matrices, and lists in R. Master vectors = master R!